

NASA Exoplanet Data Set Analysis

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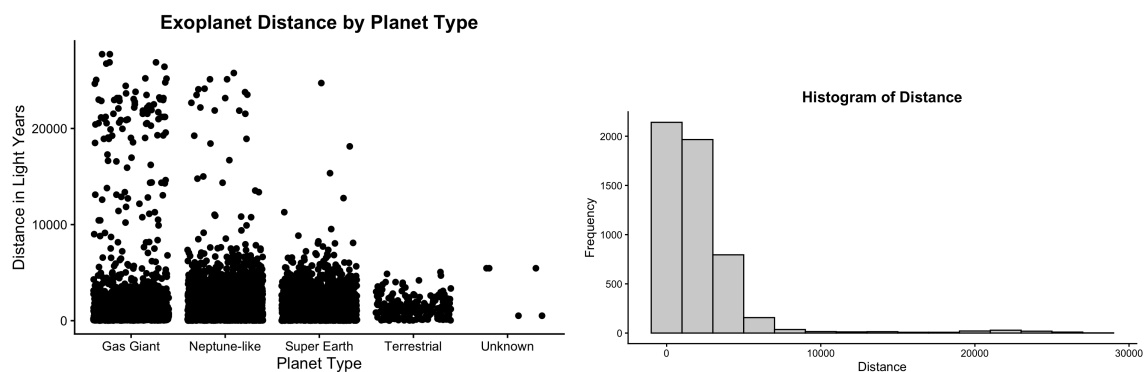
05/26/2026

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This research utilizes the data set *NASA Exoplanets*. The purpose of this data set is to archive information from the NASA exoplanet archive. An exoplanet is any planet that resides outside our solar system. This dataset includes several variables that we will observe: distance, stellar magnitude, planet type, mass wrt, mass multiplier, radius wrt, radius multiplier, and orbital radius. This paper will analyze the variables utilized in the data set. Then, three hypotheses will be formed and tested using the given information.

Distance

In this context, distance is a planet's distance from Earth in light-years. Neptune-like planets seem to be the farthest distance from our solar system. The minimum distance is 4 light-years from our solar system while the maximum is 27727 light-years away. The mean distance is 2167 light-years with a median of 1371 light-years. The first quartile is 389 light-years while the third quartile is 2779 light-years. The values for distance are right-skewed.



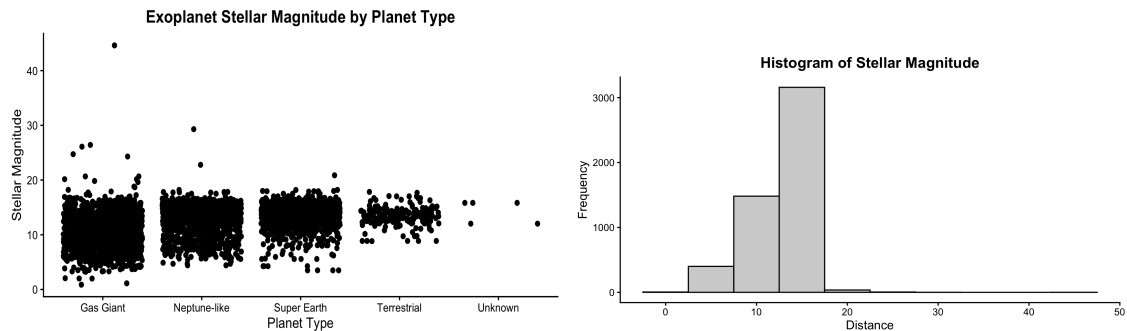
Stellar Magnitude

In this context, stellar magnitude is the brightness of a planet with a lower stellar magnitude indicating a brighter planet. The minimum stellar magnitude is 0.872 while the

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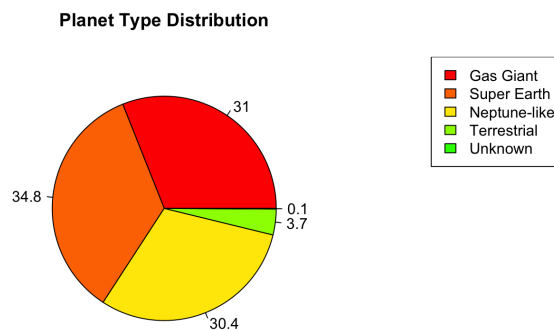
maximum is 44.610. The mean stellar magnitude is 12.684 with a median of 13.543 light-years.

The first quartile has a stellar magnitude of 10.939 while the third quartile has a stellar magnitude of 15.021. The values for stellar magnitude are overall normally distributed.



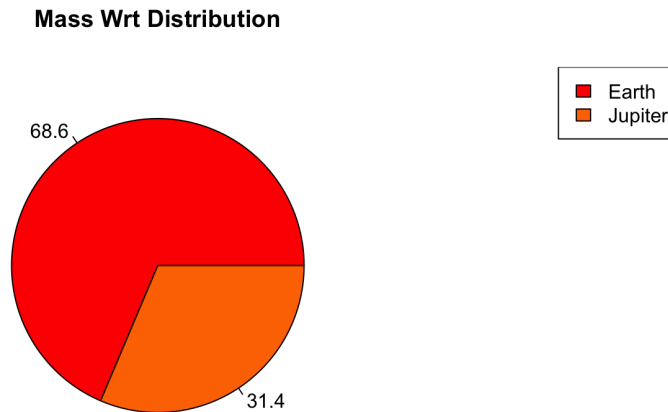
Planet Type

Planet type refers to the category of the planet a given exoplanet falls into. This particular dataset separates planets into five different types. Neptune-like planets are exoplanets that share a similar composition to that of Neptune or Uranus. Gas Giants are planets that are primarily composed of helium or hydrogen. A Super Earth is a planet that is larger than Earth but smaller than ice giants such as Neptune and Uranus. Terrestrial planets are planets composed of rock and metal with a hard surface. The final category, unknown, refers to planets that haven't been designated a specific planet type. The largest category is Neptune-like planets while the smallest is the unknown category.

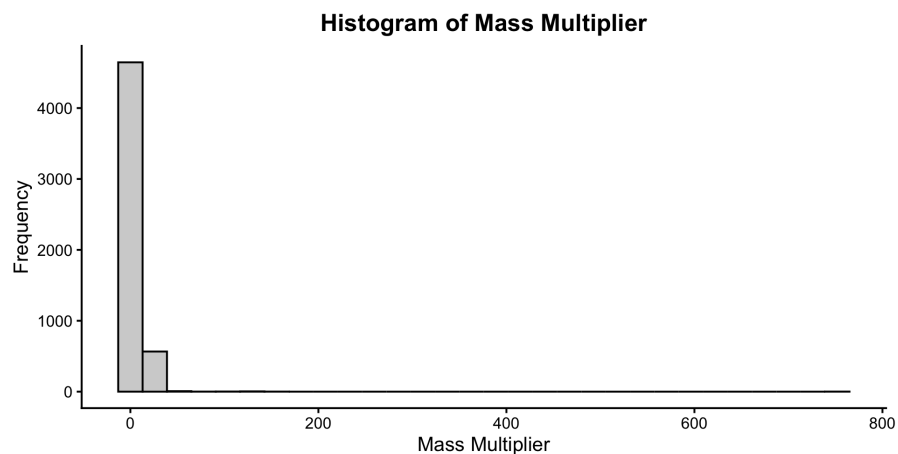


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Mass Wrt. In the context of this data set, Mass Wrt is the planet in our solar system used to compare the mass of an exoplanet. The only planets used in this category are Earth which comprises 68.6% of the composition, and Jupiter which comprises 31.4%.



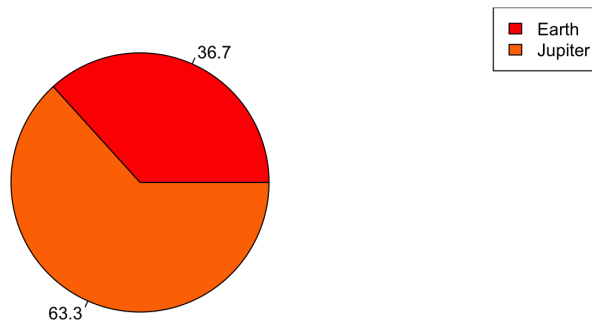
Mass multiplier. In the context of this data set, the mass multiplier is the factor in which an exoplanet is larger compared to its mass wrt, one of the planets in our solar system. The minimum mass multiplier in this data set is 0.020 with the maximum being 752.000. The first quartile is 1.804 while the third quartile is 8.000. The mean mass multiplier is 6.435 while the median is 4.170. The values for the mass multiplier are right-skewed.



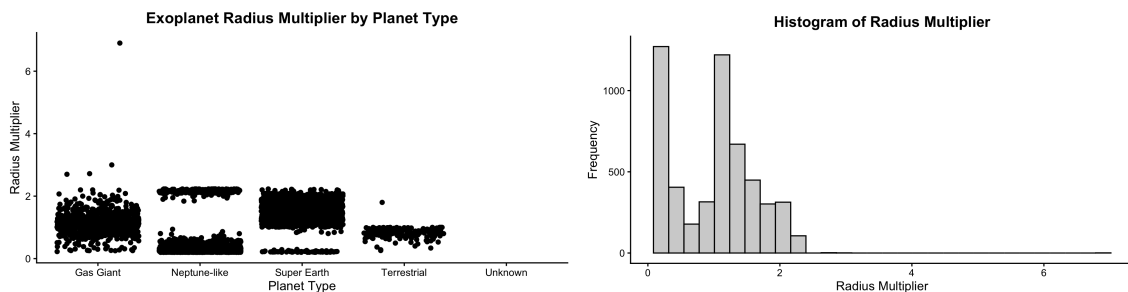
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Radius Wrt. In the context of this data set, Radius Wrt is the planet in our solar system used to compare the radius of an exoplanet. The only planets used in this category are Earth which comprises 63.3% of the compositions and Jupiter which comprises 36.7%.

Radius Wrt Distribution



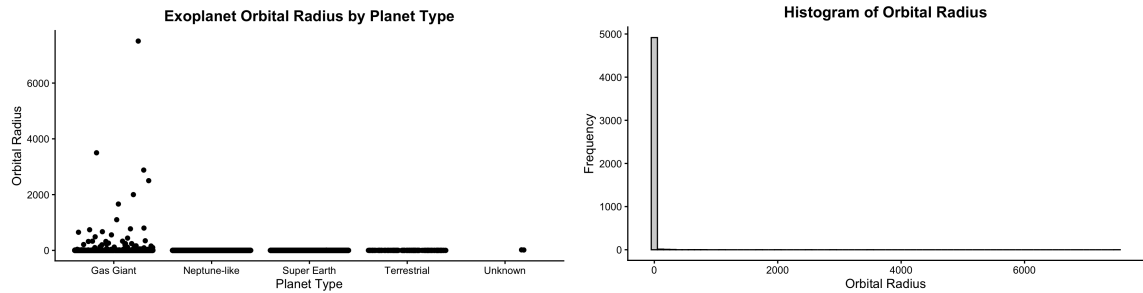
Radius Multiplier. In the context of this data set, the radius multiplier is the factor in which an exoplanet is larger compared to its radius wrt, one of the planets in our solar system. The minimum radius multiplier in this data set is 0.200 with the maximum being 6.900. The first quartile is 0.325 while the third quartile is 1.410. The mean radius multiplier is 1.015 while the median is 1.120. The values for the radius multiplier are right-skewed.



Orbital Radius. In the context of this data set, the orbital radius is the radius at which an exoplanet orbits its sun. Orbital radius is measured using Astronomical Units (AU). The minimum orbital radius is 0.004 AU while the maximum is 7506.000 AU. The first quartile is

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0.053 AU while the third quartile is 0.285 AU. The mean is 6.963 AU while the median value is 0.103 AU. The values for orbital radius are right-skewed.

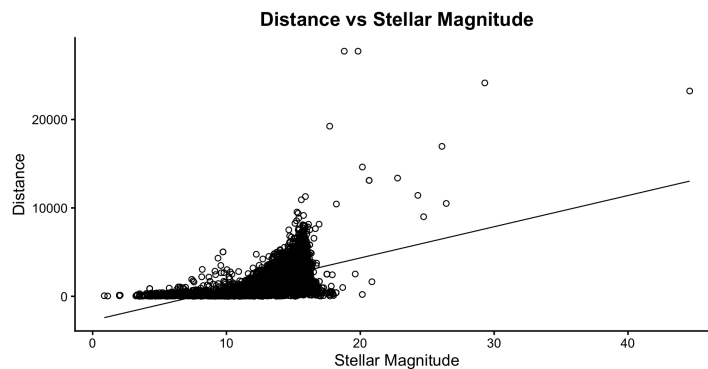


Hypothesis Testing

The following section will present three research questions based on the previously expressed data. The appropriate testing will be conducted to test the hypothesis presented.

Is there a correlation between stellar magnitude and distance?

This research question aims to observe the relationship between an exoplanet's distance from Earth and its stellar magnitude. It's of interest as by answering this question, we will be able to see if these two correlate with one another which may indicate that one is able to predict the other. I was also interested in these two variables in particular, as they weren't as right-skewed as the other numeric variables.



I hypothesize that there isn't a correlation between stellar magnitude and distance due to the graph being largely right-skewed. The methods that will be utilized to answer the question are Correlation & Simple Linear Regression. This method is appropriate as we will be comparing two numeric values. Now we will find and interpret the slope of the regression line:

```
(Intercept) ep$stellar_magnitude
-2720.8983      353.1239
```

This suggests that as the stellar magnitude increases by 1, the distance increases by 353.1 light-years. The intercept of -2720.9 serves as an extrapolation and isn't necessarily accurate. Now we will perform a correlation test. The null hypothesis states that as Stellar magnitude increases, distance is not expected to change. The alternative hypothesis states that as Stellar magnitude increases, distance is also expected to change. For this scenario, $\alpha = 0.05$.

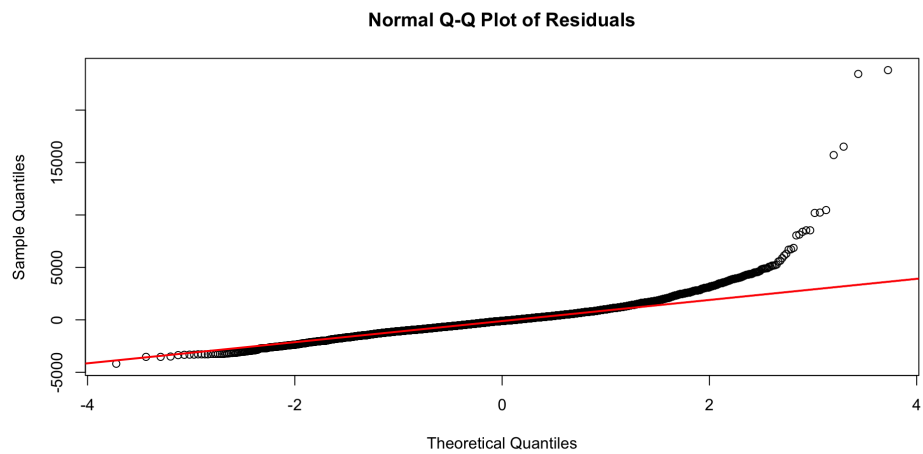
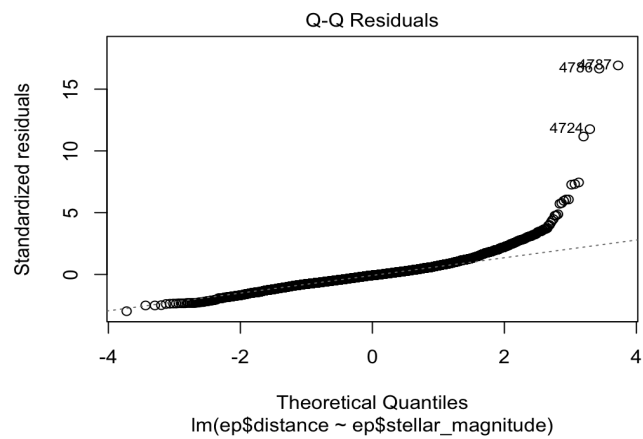
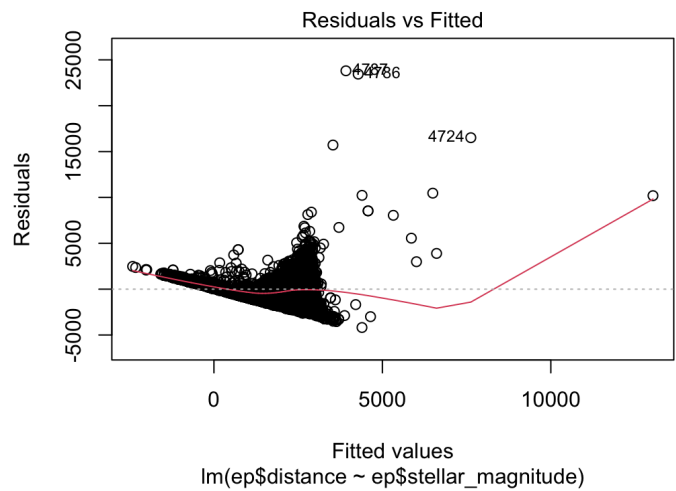
```
Pearson's product-moment correlation
```

```
data: ep$distance and ep$stellar_magnitude
t = 55.495, df = 5075, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.5971280 0.6313767
sample estimates:
      cor
0.6145418
```

```
[1] 0.3776617
```

Since the p-value is less than the alpha of $\alpha = 0.05$, we can reject the null hypothesis meaning that as Stellar magnitude increases, distance is also expected to change. 37.77% of the variation in distance is explained by its relationship with Stellar Magnitude. Now we will visually check the model's assumption.

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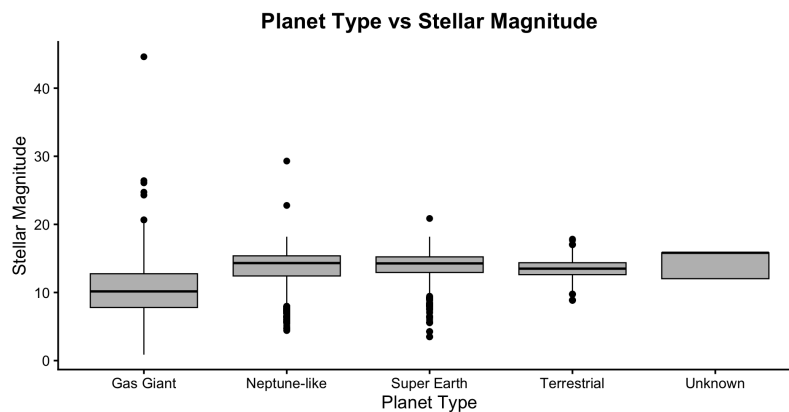
The residuals appear to be primarily centered around overall consistent variance. There are outliers but not an extreme amount. Since the residual points closely follow the line, a normality assumption is reasonable. Now we will find and interpret the residual standard error.

[1] 1408.066

This value means that we can expect on average that 68% of predicted distance values are within 1408.066 light-years of their actual distance.

Is there a relationship between stellar magnitude and planet type?

This research question aims to observe the relationship between stellar magnitude means and planet type. This is of interest to the data set, as we will be able to see if stellar magnitude is in any way related to what category an exoplanet falls into.

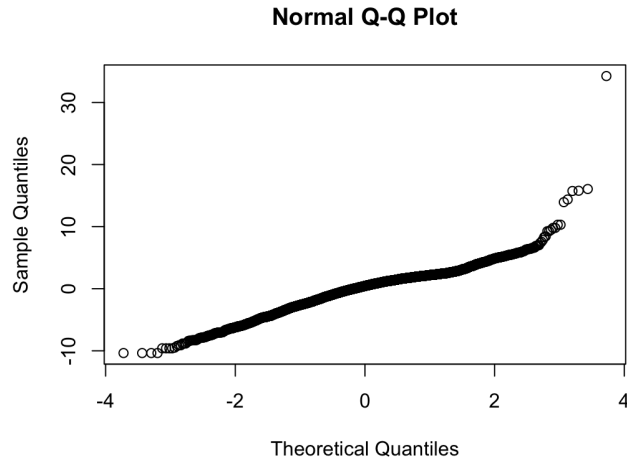


I hypothesize that stellar magnitude does have a relationship with planet type. In the box plot presented above, it appears there is some difference between each planet type's amount of stellar magnitude. The method that will be utilized to answer the question is ANOVA. This method is appropriate as we will be comparing the means of planet types to see if there's a significant difference. In this case, the null hypothesis is that there is no significant difference between

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stellar magnitude means across planet types. The alternative hypothesis states that there is a significant difference between stellar magnitude means across planet types. For this scenario, $\alpha = 0.05$. The following is the conducted ANOVA test and Q-Q plot:

```
              Df Sum Sq Mean Sq F value Pr(>F)
ep$planet_type  4  11731   2932.7   398.6 <2e-16 ***
Residuals      5084  37404     7.4
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
161 observations deleted due to missingness
```



The Normal Q-Q Plot appears to indicate an overall normal distribution. Meanwhile, the ANOVA test has a p-value below the alpha of 0.05. This means that we can reject the null hypothesis, meaning there is a significant difference between stellar magnitude means. Now we will determine where this difference occurs by using Bonferroni's method.

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Pairwise comparisons using t tests with pooled SD

data: ep\$stellar_magnitude and ep\$planet_type

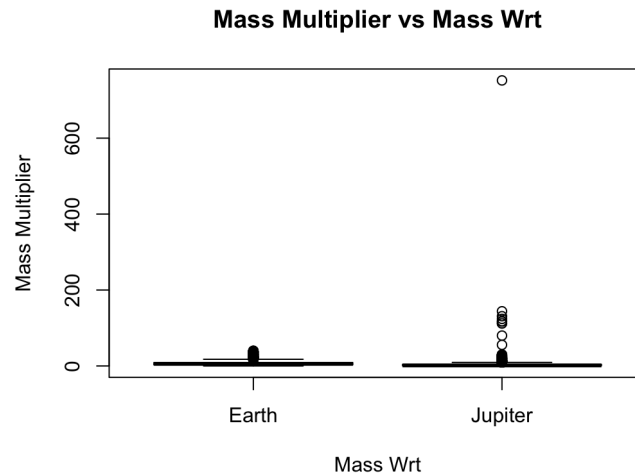
	Gas Giant	Neptune-like	Super Earth	Terrestrial
Neptune-like	<2e-16	-	-	-
Super Earth	<2e-16	0.0036	-	-
Terrestrial	<2e-16	1.0000	0.4922	-
Unknown	0.0116	1.0000	1.0000	1.0000

P value adjustment method: bonferroni

It appears that the following planet type pairs display a significant difference along stellar magnitude means: Neptune-like and Gas Giant, Super-Earth and Gas Giant, Terrestrial and Gas-Giant, Neptune-like and Super Earth, and Unknown and Neptune-like. To conclude the findings of this research question, there is a significant difference between the stellar magnitude means based on planet type with these differences occurring at Neptune-like and Gas Giant, Super-Earth and Gas Giant, Terrestrial and Gas-Giant, Neptune-like and Super Earth, and Unknown and Neptune-like.

Is there a relationship between an exoplanet's Mass Multiplier and its Mass Wrt?

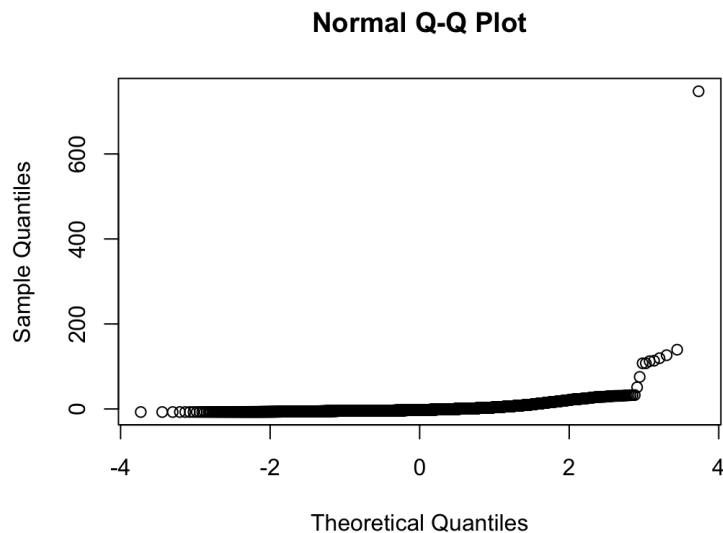
This research question aims to observe the relationship between Mass Multiplier and Mass Wrt. This is of interest to the data set, as we will be able to see if Mass Multiplier is impacted by what Mass Wrt is used for comparison, whether it be Earth or Jupiter.



I hypothesize that Mass Multiplier doesn't have a relationship with Mass Wrt. This is due to the box showing similar results for the Mass Multiplier of Earth and Jupiter except for a few outliers. The method that will be utilized to answer the question is ANOVA. This method is appropriate as we will be comparing the Multiplier Mass means to see if there's a significant difference among Mass Wrt. In this case, the null hypothesis is that there isn't a significant difference between mass multiplier means across Mass Wrt. The alternative hypothesis states that there is a significant difference between mass multiplier means across Mass Wrt. For this scenario, $\alpha = 0.05$. The following is the conducted ANOVA test and Q-Q plot:

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```
          Df Sum Sq Mean Sq F value    Pr(>F)
ep$mass_wrt  1   8427     8427   50.55 1.32e-12 ***
Residuals 5225 871065     167
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
23 observations deleted due to missingness
```



The Normal Q-Q Plot appears to indicate an overall normal distribution. Meanwhile, the ANOVA test has a p-value below the alpha of 0.05. This means that we can reject the null hypothesis, meaning there is a significant difference between Mass multiplier means. Now we will determine where this difference occurs by using Tukey's method.

```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = ep$mass_multiplier ~ ep$mass_wrt)
```

```
$`ep$mass_wrt`
          diff      lwr      upr p adj
Jupiter-Earth -2.735914 -3.490307 -1.981521    0
```

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It appears that Jupiter and Earth display a significant difference in mass multiplier means due to having a p-value of 0. To conclude the findings of this research question, there is a significant difference between the mass multiplier means for the Mass Wrts Earth and Jupiter.

References

<https://www.kaggle.com/datasets/adityamishraml/nasaexoplanets>